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Annexure3b- Complete filing

INVENTION DISCLOSURE FORM

1. TITLE: Self-Healing Aircraft Wing Using Smart Materials

2. INTERNAL INVENTOR(S)/STUDENTS(S): All fields in this column are mandatory to be filled.

3. DESCRIPTION OF THE INVENTION:

Our invention is an aircraft wing that can repair itself in case any cracks appear in it without having to be opened up by a mechanic every time. Some smart material layers were added onto the aircraft wing and react instantly when any cracks appear.

A. PROBLEM ADDRESSED BY THE INVENTION: Aircraft wings are subject to heavy-duty operations, high pressure, temperature variations, and vibration when in operation. A crack can occur in a plane's wings made of composite structures, and it may go unnoticed during the inspections, leading to the deterioration of the airplane's wing structure, posing a risk to the aircraft. The crack would need to be repaired, necessitating the grounding of the aircraft during the maintenance process.

B. OBJECTIVE OF THE INVENTION

To build a wing structure that can detect internal cracks on its own using embedded smart materials.

To make the wing repair small cracks automatically, without needing the aircraft to be taken out of service.

To increase the working life of the wing and reduce the frequency of manual structural inspection.

C. STATE OF THE ART/RESEARCH GAP/NOVELTY: Describe how your invention fulfils the research gap.

D. DETAILED DESCRIPTION:

The present invention relates to a self-healing aircraft wing structure incorporating smart materials, embedded structural health monitoring elements, localized damage detection mechanisms, and an autonomous repair mechanism. The proposed aircraft wing is designed to detect the formation of micro-cracks or localized structural damage within the composite structure and initiate a self-healing process without requiring dismantling of the aircraft wing or immediate manual maintenance intervention.

D.1 Artificial Intelligence Models Used:

The proposed invention utilizes multiple Artificial Intelligence (AI) and Machine Learning (ML) models for crack detection, assessment of resulting damage, localization of damaged areas, and verification of the healing process. The Convolutional Neural Network is responsible for analyzing images captured by the embedded vision sensors and identifying cracks. The Long Short-Term Memory model is applied to evaluate the patterns of

collected data and predict the development of cracking. An Autoencoder anomaly detection model is used for structural damage identification by comparing the acquired measurements using various sensors. The Reinforcement Learning model allows the wing to heal itself by constantly learning the optimal set of actions for healing activation while minimizing the energy consumption for the healing process. The described AI models enable an autonomous system that can assess, analyze, and heal cracks in an aircraft wing.

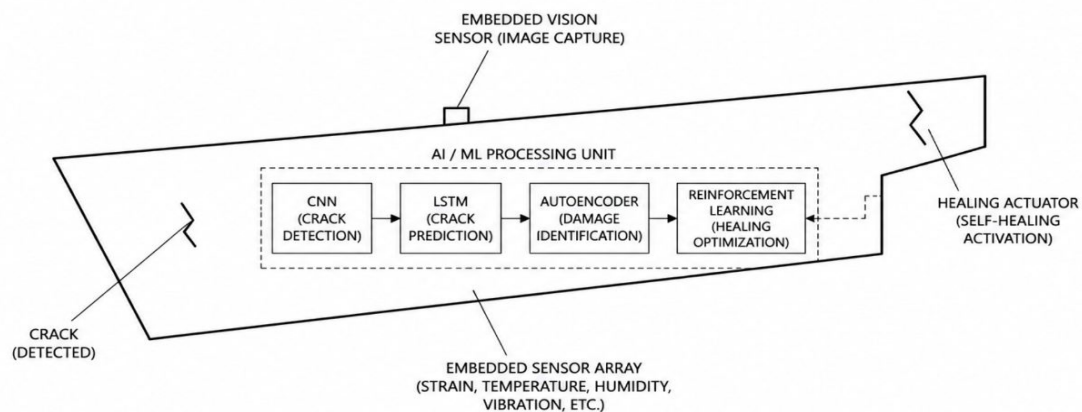


FIG. 1

Figure 1 illustrates the overall structural architecture of the proposed self-healing aircraft wing, including the outer composite layer, structural composite layers, embedded sensing layer, smart healing material layer, conductive heating elements, and control unit.

D.2 Hardware Components:

The suggested prototype includes a multilayered aircraft wing made of fiber optics sensors, piezoelectric sensors, strain gauges, acoustic emission sensors, heating elements, carbon nanotube smart healing material, microcontroller, industrial processor, wireless communication module, power supply, and storage unit. These hardware components enable crack sensing and monitoring, localized heating, self-healing initiation, healing progress observation, and data storage.

D.3 Software Components:

The framework used for the development of this software includes Python for the coding of AI algorithms, TensorFlow or PyTorch for deep learning, OpenCV for the processing of crack images, MATLAB for simulations and validation of results, Scikit-learn for implementing predictive models, Embedded C for microcontroller programming, and an IoT dashboard for the visualization of monitoring data including crack status, healing progress, and maintenance history.

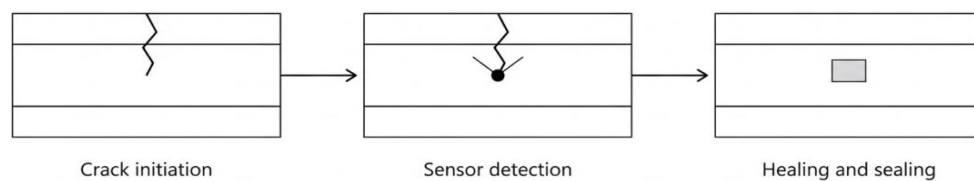


FIG. 3

Figure 2 illustrates the autonomous self-healing sequence in which a crack is detected by embedded sensors, localized heating is activated, and the smart healing material repairs the damaged region.

D.4 System Workflow:

The process may begin with monitoring the airplane wing structure for possible flaws using the sensors and the images captured by the microchips. At this stage, the artificial intelligence would be used to identify irregularities and analyze the cracks' depth and reach. If the damage is found beyond the irreplaceable size, the control unit would initiate heating of the conductor around the affected area to begin the self-healing mode. Finally, the self-healing material's structure would be observed for any flaws in the wings using the sensors and be recorded in the maintenance log if needed. In the end, the airplane would continue the flight with a healthy wing structure.

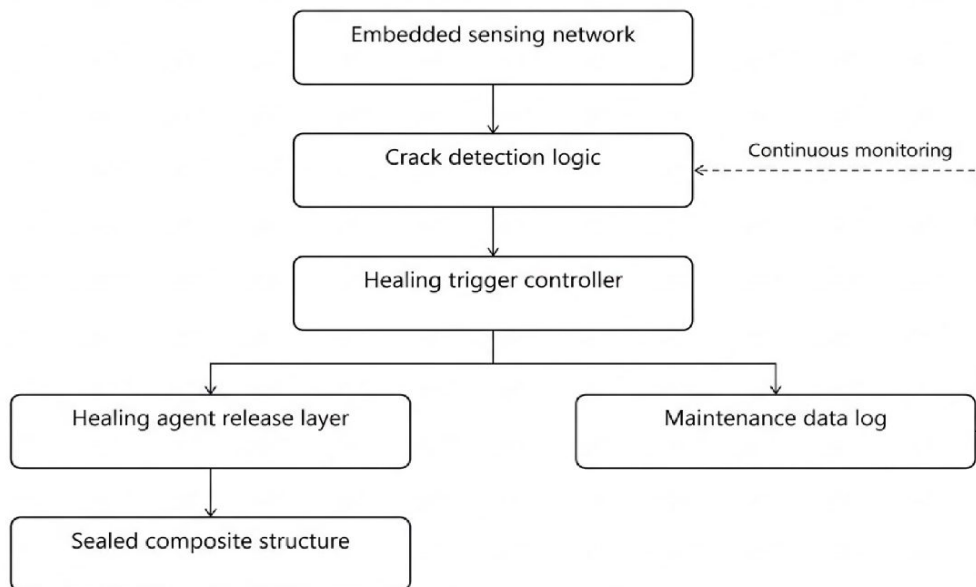


FIG. 4

D.5 Estimated Prototype Development Cost:

D.6 Industrial Applications:

The proposed self-healing aircraft wing may be implemented in commercial passenger aircraft, military aircraft, UAVs, helicopters, cargo aircraft, spacecraft, or other aircraft or structures requiring structural health monitoring. The invention may also find application in wind turbine blades, marine structures, railroad components, spacecraft structures, or other structures or materials requiring enhanced structural health and self-repair capabilities.

D.7 Integrated Health Management and Autonomous Self-Healing Mechanism

The present invention discloses a multi-layer self-healing aircraft wing comprising a plurality of structural composite layers, an embedded sensing layer, a smart healing material layer, conductive heating elements and optionally a control and monitoring unit. The structural composite layers provide the mechanical strength, rigidity, aerodynamical stability and load-bearing capability required for aircraft operation. The embedded sensing layer continuously monitors the structural parameters like strain, stress, vibration, acoustic emission, electrical resistance and conductivity for the detection of micro cracks or localized structural damages. Upon detection of damage, the monitoring unit activates localized Joule heating through conductive elements placed close to the damaged region. The thermal energy produced triggers reversible chemical reactions in intrinsic self-healing polymers, dynamic molecular networks, thermoplastic healing materials, or hybrid smart

materials to fix the damaged site without deconstructing the aircraft wing. The sensing and healing components are strategically placed at fatigue-prone locations such as the wing root, leading edge, trailing edge, joints and structural members to allow for rapid damage localization and repeatable autonomous repair. The monitoring system stores all damage events, the location of the cracks, the status of healing, the healing cycle and the structural status in a digital maintenance history. After each healing cycle, the sensing network checks the structural integrity of the healed region and, if necessary, automatically initiates another healing cycle or generates a maintenance alert. The stored structural health information can be transmitted to the onboard aircraft systems, cockpit display, ground maintenance stations or remote diagnostic sites through wired or wireless communication networks. The proposed invention thus provides an integrated structural health management framework capable of autonomously detecting damage, localizing damage, repeatedly self-healing, verifying healing, logging maintenance data and continuously monitoring structural health, thereby improving the safety of the aircraft, reducing maintenance downtime and extending the service life of the aircraft wing.

WORKING OF THE INVENTION

In the case of normal functioning of the aircraft, the integrated sensing network constantly or periodically measures the structural status of the wing of the aircraft.

When a micro-crack occurs, the crack causes changes in the structure that can be detected by sensing elements.

The sensing network detects the abnormal structural condition and the area that is possibly affected by the damage.

The system analyzes whether the abnormal structural condition detected by the sensing network meets a pre-defined threshold for activating the self-healing process.

In the case when the abnormal condition meets the defined threshold, the healing process related to the particular damaged area is initiated.

Specific electrical, thermal, mechanical, or material-responsive activation triggers the process of self-healing.

The intelligent healing material seals and heals the damaged part of the aircraft.

Once the process of healing is completed, the repaired area is left to stabilize.

Post-healing analysis is conducted by the sensing network to detect whether the abnormal condition has healed down to the acceptable level.

The information about damage occurrence, healing process, location of the repaired area, and number of healing cycles is recorded.

In case the structural features of the healed section remain abnormal, the system issues a warning for maintenance purposes.

Therefore, the current invention provides a method of making a self-healing aircraft wing that will be able to identify structural damage, conduct local healing, confirm the healing process, keep records of healing activities, and monitor structural health.

E. RESULTS AND ADVANTAGES:

Expected Results: The wing is expected to detect and seal small cracks within a short time of their forming, reducing the chances of the damage spreading. It should also reduce how often the aircraft needs a full structural inspection for hidden micro-cracks.

Advantages:

Repairs small damage without stopping the regular use of the aircraft

Reduces long-term maintenance cost and inspection downtime

Extends the working life of the wing structure

Lowers the safety risk that comes from undetected internal cracks

F. EXPANSION: The same smart healing layer can later be used in other parts of the aircraft body, such as the fuselage or tail section, and can also be adapted for use in wind turbine blades, which face similar stress and crack problems.

G. WORKING PROTOTYPE/ FORMULATION/ DESIGN/COMPOSITION:

The working prototype is under development.

H. EXISTING DATA: Any clinical or comparative data necessary to support your invention. (Comparative). NA

4. USE AND DISCLOSURE (IMPORTANT): Please answer the following questions:

5. Provide links and dates for such actions if the information has been made public (Google, research papers, YouTube videos, etc.) before sharing with us. NA

6. Provide the terms and conditions of the MOU, as well as whether the work is done in collaboration within or outside the university (any industry, other universities, or any other entity). NA

7. Potential Chances of Commercialization. Yes

8. List of companies that can be contacted for commercialization, along with the website link.

Airbus

Boeing

Lockheed Martin

Northrop Grumman

HAL (Hindustan Aeronautics Limited)

Collins Aerospace

Safran

GE Aerospace

Rolls-Royce Aerospace

Spirit AeroSystems

9. Any basic patent that has been used and for which we need to pay royalties to them. NA

10. FILING OPTIONS: Please indicate the level of your work that can be considered for provisional/complete/PCT filings. Complete

11. KEYWORDS: Self-healing aircraft wing, smart materials, structural health monitoring (SHM), composite materials, autonomous crack repair, embedded sensors, artificial intelligence, machine learning, crack detection, structural damage monitoring, conductive heating, Joules heating, dynamic polymer network, thermoplastic healing, carbon nanotubes, fiber optic sensors, piezoelectric sensors, aerospace composites, aircraft maintenance, predictive maintenance, intelligent aerospace structural, repeatable self healing, aerospace safety, digital structural health management.

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Sr. No.	Patent ID	Abstract	Research Gap	Novelty
I.	US6518330B2	This patent puts tiny capsules with a healing agent into the concrete to seal the cracks.	This system has a healing agent that only goes to the cracks and not the rest of the concrete.	It is a self-healing concrete that can heal itself.
II.	US20090036568A1	This patent's material heals itself when it is heated.	This self-healing material is made of a polymer that can be heated to activate the healing process.	The material can be heated to activate the healing process.
III.	US20130189888A1	This patent builds tiny tubes into the concrete that can deliver healing agents.	Building these tubes is a challenge because they need to be strong enough to hold the concrete together.	The tubes can deliver healing agents to the cracks.
IV.	US20230036334A1	It is an aerospace epoxy that can self-heal.	The patent is for a self-healing epoxy that can be used in aerospace applications.	The epoxy can self-heal when it is heated.
V.	US20160167811A1	A tri-layered self-healing concrete with a reactive core.	Concrete with a reactive core that can heal itself when it is heated.	The concrete can self-heal when it is heated.
VI.	US9150721B2	A combination of the thermally activated epoxy and the microcapsules.	The combination of the thermally activated epoxy and the microcapsules is a novel approach to self-healing concrete.	The combination of the thermally activated epoxy and the microcapsules is a novel approach to self-healing concrete.

Component	Estimated Cost (INR)
Composite Wing Material	■ 5,00,000
Smart Self-Healing Material	■ 3,50,000
Fiber Optic & Piezo Sensors	■ 2,50,000
Conductive Heating Elements	■ 1,20,000
Embedded Controller & Processor	■ 1,50,000
AI Software Development	■ 3,00,000
Laboratory Testing & Validation	■ 2,30,000
Total Estimated Cost	■ 18-20 Lakhs

Module	AI Model	Purpose
Crack Detection	CNN / Vision Transformer (ViT)	Detect and classify surface cracks
Structural Health Monitoring	LSTM	Analyze vibration, strain and acoustic signals
Damage Prediction	Random Forest / Gradient Boosting	Predict crack growth and structural degradation
Anomaly Detection	Autoencoder	Detect abnormal structural behaviour
Healing Optimization	Reinforcement Learning (Q-Learning / Deep Q-Network)	Optimize healing activation
Continuous Learning	Online Learning	Improve healing performance from previous repair cycles
Maintenance Prediction	Predictive Analytics	Estimate maintenance requirements and remaining useful life

Have you described or shown your invention to anyone or in any conference?	YES () NO (✓)
Have you made any attempts to commercialize your invention (for example, have you approached any companies about purchasing or manufacturing rights)?	YES () NO (✓)
Has your invention been described in any printed publication, or any other form of media, (such as the Internet)?	YES () NO (✓)
Do you have any collaboration with any other institute or organization on the same?	YES () NO (✓) (if name and other details.)
Name of Regulatory body or any other approval authority, if required.	YES () NO (✓)